

UNSUPERVISED LEARNING — PR 1

Final Conclusion & Business Interpretation

Project Overview

The project applied K-Means Clustering, Agglomerative Hierarchical Clustering, and DBSCAN to the Mall Customer Segmentation Dataset. The objective was to identify meaningful customer groups using income and spending behaviour.

Data Preparation

- The dataset contains 200 customer records and 5 original columns.
- CustomerID was removed because it is an identifier.
- Annual Income (k\$) was renamed to Annual_Income and Spending Score (1-100) to Spending_Score.
- Gender was encoded using LabelEncoder: Female = 0 and Male = 1.
- Age, Annual_Income, and Spending_Score were standardized using StandardScaler.
- Annual_Income and Spending_Score were used as the primary two-feature clustering space.

K-Means Conclusion

The Elbow Method and Silhouette Score supported a five-cluster solution. The final K-Means model used $k = 5$, $\text{random_state} = 42$, and $n_init = 10$. The approximate Silhouette Score was 0.555.

Customer Segment Interpretation

Segment	Profile	Suggested Business Action
Cluster 0	Moderate income / moderate spending	General promotions and personalized offers
Cluster 1	High income / high spending	VIP rewards, premium products, exclusive offers
Cluster 2	Low income / high spending	Discounts, bundles, affordable premium products
Cluster 3	High income / low spending	Personalized recommendations and incentives
Cluster 4	Low income / low spending	Budget products, value bundles, entry-level rewards

Agglomerative Hierarchical Clustering Conclusion

Ward-linkage hierarchical clustering was fitted with five clusters. The dendrogram supported a five-cluster cut. Its profiles were broadly consistent with K-Means, although some boundary customers were assigned differently because hierarchical clustering uses a different clustering strategy.

DBSCAN Conclusion

DBSCAN was tuned using the 4-nearest-neighbour distance plot and the required parameter grid. The selected configuration was $\text{eps} = 0.4$ and $\text{min_samples} = 3$. It produced 4 clusters and 10 noise observations. DBSCAN therefore provided a density-based view and explicitly identified isolated observations.

Algorithm Comparison

Algorithm	Clusters	Silhouette	Davies-Bouldin	Calinski-Harabasz
K-Means	5	0.555	0.572	248.649
Hierarchical	5	0.554	0.578	244.410
DBSCAN	4	0.395	0.601	97.172

For DBSCAN, noise observations were excluded from metric calculations. K-Means and Hierarchical Clustering produced very similar five-cluster structures. K-Means provides the clearest and most interpretable segmentation for this project.

Final Business Conclusion

The analysis reveals distinct customer groups based on annual income and spending behaviour. These segments can support targeted marketing instead of applying one common strategy to all customers.

High-income/high-spending customers can receive premium and loyalty-oriented offers. Low-income/high-spending customers can receive value-oriented promotions and bundles. High-income/low-spending customers can receive personalized recommendations and incentives. Low-income/low-spending customers can receive budget-friendly offers. Moderate-income/moderate-spending customers can be developed through cross-selling and personalized rewards.

Overall Conclusion

The project successfully completed the required unsupervised learning workflow. K-Means produced a clear five-segment solution, Hierarchical Clustering provided a useful dendrogram-based view, and DBSCAN provided density-based clustering with explicit noise detection.

Based on the combined clustering structure, evaluation metrics, and interpretability, K-Means is used as the primary segmentation approach, while Hierarchical Clustering and DBSCAN provide complementary analytical perspectives.